

Formal Rules and Their Bite: The Role of Private Communication*

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We study how private communication can render constraints on formal decision rules ineffective for restricting implementability. Unlike standard mechanism-design analyses that impose constraints on the entire mechanism, we assume that only an externally auditable formal rule mapping reports into outcomes is constrained, while the principal's private recommendations are off-the-record and unconstrained. We provide a sufficient condition for implementability based on an almost-sure obedience requirement. Fairness constraints such as symmetry lose their bite: once off-the-record recommendations are available, any incentive-compatible social choice function is implementable, even under dominant-strategy requirements. Predictability constraints, by contrast, retain bite: strict monotonicity and continuity preclude almost-sure obedience. Our results suggest that when predictability constraints do not distort the target social choice function, they may also restore bite to fairness requirements.

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1. Introduction

In organizations and institutions, formal decision rules are often externally auditable and subject to constraints. Performance evaluation systems in firms, procurement rules in public agencies, and credit-scoring standards in financial institutions are all documented in writing, so that regulators and auditors can later check their content. A common design tool for ensuring fair decision-making is to impose constraints—such as equal treatment, non-discrimination, and transparency—on these formal rules. However, such formal rules often fail to fully determine which outcomes are implemented, in part because principals can communicate privately with agents. Off-the-record advice, informal coaching, and guidance exchanged prior to formal reporting are commonplace but remain largely outside the scope of the external audit.

Consider, for instance, a firm’s performance evaluation system. The formal evaluation criteria are documented, disclosed to all employees, and subject to audit, so the system appears symmetric on paper. At the same time, a supervisor may provide off-the-record coaching to a favored employee on how to write the self-evaluation in a way that matches the criteria that happen to carry weight in a given review cycle. Employees who do not receive such guidance face the same formal rule but are unlikely to frame the self-evaluation in the “right” way, and consequently receive lower ratings. The formal rule is symmetric; the implemented outcome is not.

This raises a fundamental question: Even if one imposes stringent constraints on formal rules, can off-the-record private communication render those constraints irrelevant for implementability? If so, which constraints remain effective when advice and guidance can be exchanged without external scrutiny? This question is practically important and therefore worth addressing; however, it resists analysis within the standard mechanism-design framework. This is because, in the canonical formulation, the mechanism is treated as a single object, and constraints are imposed on the mechanism as a whole. For example, [Azrieli and Jain \(2018\)](#) show that when symmetry is imposed on the entire mechanism, dominant-strategy implementation is possible only for symmetric social choice functions. However, that approach imposes symmetry on the mechanism as a whole, whereas in many real institutions only the officially documented decision rule is externally auditable and subject to regulation, while off-the-record

advice and informal guidance remain largely beyond the scope of external scrutiny.

This paper explicitly captures the institutional distinction between formal rules and private communication. In our model, formal rules are externally auditable and must satisfy the prescribed constraints. By contrast, private communication from the decision maker (the principal) to agents is off-the-record and is not subject to any constraints. With constraints applying only to the externally auditable formal rule, we analyze which constraints on formal rules lose their bite for implementability once off-the-record private communication is available.

Our main finding is that whether a constraint remains effective hinges on what kind of restriction it imposes. We focus on two broad classes: fairness and predictability constraints. Fairness constraints, epitomized by symmetry, require the formal rule to treat agents symmetrically, but they leave considerable freedom in how outcomes respond to changes in reports. By contrast, predictability constraints, exemplified by strict monotonicity and continuity, require that changes in reports be systematically and stably linked to changes in outcomes, thereby disciplining the report-outcome relationship itself.

The results yield a clear contrast. Fairness constraints lose their bite for implementability once off-the-record private communication is available. Even when the formal rule is required to be symmetric, private communication allows the implementation of any dominant-strategy incentive-compatible social choice function whose outcomes lie in the class of non-punishment outcomes covered by our construction. This sharply contrasts with [Azrieli and Jain \(2018\)](#), where under dominant strategies imposing symmetry on the entire mechanism restricts implementation to symmetric social choice functions. In our setting, once the externally auditable formal rule is separated from off-the-record private recommendations and symmetry is imposed only on the former, this restriction disappears. Predictability constraints, by contrast, retain real bite. Under strict monotonicity, off-the-record private communication does not remove the restriction. For continuity, the situation is more delicate: continuity still has bite in that it rules out exact obedience and hence exact implementation, although in some environments continuous rules can still satisfy an approximate obedience property. Whether such an approximation is sufficient depends on the structure of outcomes, and implementability is not guaranteed in general.

The intuition behind these results is as follows: When private communication is permitted, the formal rule can be equipped with a “password-like” verification structure: the intended outcome is realized only when the agent’s report matches the privately recommended “correct” report, while any other report triggers a punishment outcome. Symmetry is compatible with such a construction, because verification can be embedded in a way that does not depend on agents’ identities, thereby preserving symmetry. By contrast, monotonicity requires outcomes to improve when an agent reports a higher value along the relevant dimension. As a result, even if the agent departs from the recommendation, reporting a slightly higher value in that dimension can avoid punishment. Continuity likewise rules out “punishment-for-any-deviation” designs, because it forbids sharp outcome changes in response to arbitrarily small changes in reports. In short, an institution’s vulnerability to private communication is governed by how strongly the imposed constraint forces the input-output relationship of the formal rule to be predictable.

These findings have implications for institutional design. Even if fairness constraints are normatively desirable, imposing them on formal rules does not, by itself, meaningfully restrict implemented outcomes as long as off-the-record private communication is present. To achieve fair decision-making in practice, fairness constraints must be complemented by predictability constraints, such as monotonicity with respect to transparent performance indices and stability of evaluations under changes in reports, that discipline how reports map into outcomes and must be embedded in the formal rule itself.

Returning to the personnel-evaluation example, the criteria emphasized in a given review cycle correspond to the random verification label in our construction, and the supervisor’s off-the-record coaching corresponds to the recommendation that reveals that label. An employee who does not receive coaching cannot frame the self-evaluation in the way that best aligns with the relevant criteria, just as an agent who does not receive the recommendation cannot match the password. If, however, the evaluation system is required to be monotonic, meaning that a higher value of an observable performance index must lead to a weakly better evaluation, then an employee can improve the outcome simply by raising that index, regardless of coaching. The private channel loses its power because predictability gives every employee a reliable route to

a better outcome.

The rest of the paper is organized as follows. Section 2 summarizes the related literature. Section 3 sets up our model. Section 4 introduces the almost-sure obedience property and defines a mechanism. Section 5 defines implementability under a set of admissible formal rules and studies which constraints lose their bite and which retain it under private communication. Section 6 discusses extensions and limitations.

2. Related literature

We relate our analysis to several strands of literature and explain how our approach differs from existing frameworks.¹

Symmetric implementation. The closest reference is [Azrieli and Jain \(2018\)](#), who study implementation by *symmetric* mechanisms. A key motivation for this line of work is that symmetry of the rule does not by itself guarantee symmetric outcomes: in an auction environment, [Deb and Pai \(2017\)](#) show that discrimination can arise even under symmetric auction rules. Against this background, [Azrieli and Jain \(2018\)](#) prove that under Bayesian Nash equilibrium, even asymmetric SCFs that satisfy BIC can be implemented by symmetric mechanisms.² By contrast, under dominant-strategy equilibrium, symmetry becomes restrictive: only symmetric social choice functions are implementable by symmetric mechanisms. [Chen and Knyazev \(2023\)](#) revisit this restriction and show that under a weaker notion of dominance (M-DSE), some asymmetric rules can be implemented by symmetric mechanisms.³ Our departure is to separate the auditable formal rule from off-the-record private communication and to impose symmetry only on the former. As a result, asymmetric outcomes can be implemented: in our constructions, private advice may break symmetry even though the auditable formal rule remains symmetric.

¹We use standard abbreviations throughout: social choice function (SCF), Bayesian incentive compatibility (BIC), and dominant-strategy incentive compatibility (DSIC).

²This conclusion is environment dependent and can fail, for example, with correlated types or interdependent values; see [Azrieli and Jain \(2018\)](#).

³[Azrieli and Jain \(2018\)](#) adopt a strict version of weak dominance (strict for at least one profile of others' messages). Under the weaker notion that never requires strict inequality, [Chen and Knyazev \(2023\)](#) exhibit asymmetric rules implementable by symmetric mechanisms.

Randomization in mechanism design. At the level of mechanism construction, our approach is closest to [Abreu and Matsushima \(1992\)](#): both use randomization to embed a verification-like test that discourages deviations. There are two key differences. First, we obtain *exact* implementation under dominant-strategy incentive compatibility, whereas [Abreu and Matsushima \(1992\)](#) study *virtual* implementation under iteratively undominated (rationalizable) play. Second, our motivation comes from a regulatory/institutional distinction: constraints are imposed only on the externally auditable formal rule, while off-the-record private advice is unrestricted. This separation is not present in the standard virtual-implementation framework. In dominant-strategy settings, even virtual implementation requires the target SCF to be DSIC ([Tian, 1997](#)); our contribution is to identify which constraints on the formal rule are compatible with exact implementation once private communication is available.

BCE and communication equilibrium. Our two-layer construction can be interpreted within the framework of Bayes correlated equilibrium (BCE) and communication equilibrium. The principal acts as a mediator who privately recommends reports, and almost-sure obedience corresponds to the standard BCE obedience constraint when actions are interpreted as reports to the formal rule ([Forges, 1986, 1990](#); [Bergemann and Morris, 2016a,b, 2019](#)).

Our novelty is institutional: constraints are imposed only on the externally auditable formal rule, not on the off-the-record private recommendations. Given this asymmetry in external auditability between formal rules and private communication, we show that fairness constraints lose their bite even under dominant strategies when the target SCF is DSIC, whereas predictability constraints such as strict monotonicity and continuity retain bite. This fairness-predictability distinction is not captured in standard BCE analyses, which typically treat the mechanism as a single object rather than separating an externally auditable rule from off-the-record advice. Moreover, dominant-strategy obedience is not the usual focus in the BCE framework; our results strengthen obedience by making it belief-independent: following the recommendation is optimal regardless of beliefs about others' recommendations.

Continuity. Our treatment of continuity differs in both object and implication from that underlying *continuous implementation* in [Oury and Tercieux \(2012\)](#). They require continuity of

the *equilibrium-induced outcome map* and derive monotonicity-type restrictions for rationalizable implementation. By contrast, we impose continuity only on the formal rule and show that continuity still has bite even when private communication is allowed. In some environments, such as when the outcome set is convex and worst outcomes are common, continuous rules can nevertheless satisfy an approximate obedience property.

3. Model

The model features a principal and n agents. Let $N = \{1, 2, \dots, n\}$ denote the set of agents, and write $i = 0$ for the principal. Players derive utility from the profile of types and the implemented outcome. Let $\mathcal{T} = T^n$ be the set of type profiles and let $p \in \Delta(\mathcal{T})$ be a commonly known prior over $\mathcal{T}$. We assume that T is a standard Borel space with $|T| = |\mathbb{R}|$. Each $t_i \in T$ is privately observed by agent i . Let $\mathcal{O}$ be the set of outcomes; write each outcome as an n -tuple $o = (o_1, \dots, o_n)$. For each $i \in \{0\} \cup N$, let $u_i: \mathcal{T} \times \mathcal{O} \rightarrow \mathbb{R}$ denote player i 's utility function. We assume that there exists a surjection $\varphi: \mathbb{R} \rightarrow \mathcal{O}$. We make the following assumption about punishment outcomes.

Assumption 1 (Worst and symmetric punishments). There exist outcomes $\{\underline{o}^i\}_{i \in N}$, $o^* \in \mathcal{O}$, and $\mathcal{O}^* \subseteq \mathcal{O}$ such that:

- (i) (*Individual worst*) For each $i \in N$, $t \in \mathcal{T}$, and $o \in \mathcal{O}^*$, $u_i(t, o) \geq u_i(t, \underline{o}^i)$, where $\underline{o}_j^i = \underline{o}_k^i$ for each $i \in N$ and $j, k \in N \setminus \{i\}$,
- (ii) (*Multi-deviation fallback*) o^* is symmetric (i.e., $o_i^* = o_j^*$ for all $i, j \in N$), and satisfies $u_i(t, \underline{o}^j) \geq u_i(t, o^*)$ for each $i, j \in N$ with $i \neq j$ and each $t \in \mathcal{T}$.

Example 1. Consider an assignment problem without transfers with monotone preferences. Let $n = 3$ and $\mathcal{O} = \mathcal{O}^* = \{(x_1, x_2, x_3) \in \mathbb{R}_+^3 \mid x_1 + x_2 + x_3 = 1\}$. Then, for instance, $\underline{o}^1 = (0, 1/2, 1/2)$, $\underline{o}^2 = (1/2, 0, 1/2)$, $\underline{o}^3 = (1/2, 1/2, 0)$ and $o^* = (1/3, 1/3, 1/3)$ satisfy Assumption 1.

Assumption 1 is tailored to environments in which the institution can specify punishment outcomes that are strictly worse than any on-path outcome for each agent, and in which a sym-

metric fallback outcome is available when multiple deviations occur. This structure is natural in settings with divisible allocations or sufficiently rich outcome sets, where one can choose interior target allocations and reserve extreme allocations as credible punishments, as in Example 1. By contrast, the assumption can fail in environments with tight feasibility constraints or salient outside options. In settings such as assignment or matching without transfers, “receiving nothing” or “no match” can be both a worst outcome and a feasible on-path outcome when supply is scarce, leaving no strict separation between target outcomes and worst outcomes.

This paper examines which types of constraints on *formal rules* retain their bite for implementability when *private communication* is allowed. The objective of the principal is to implement an SCF through a mechanism that combines a formal rule with private communication. We now describe the key components of the model.

Social choice functions and formal rules. We define a social choice function (SCF) as a target mapping $\mu: \mathcal{T} \rightarrow \mathcal{O}$ that specifies the target outcome as a function of the agents’ true type profile.

A *formal rule* is a mapping $\nu: \mathcal{T} \rightarrow \mathcal{O}$, which assigns an outcome to each report profile. In our model, “formal” means that the rules can be audited by external authorities and must satisfy the imposed constraints. Note that formal rules are the only objects to which the restrictions regarding auditability and constraints apply in the mechanism described later. Let $\mathcal{R} = \mathcal{O}^{\mathcal{T}}$ denote the set of all outcome rules. A subset $\mathcal{C} \subseteq \mathcal{R}$ represents the set of admissible formal rules. The principal selects a menu of formal rules $M \subseteq \mathcal{C}$ and draws a formal rule $\nu \in M$ according to a countably additive probability measure P on a Borel σ -algebra of M . The drawn formal rule ν^* is hidden from the agents.

Messages from agents. Each agent $i \in N$ is endowed with a private type $t_i \in T$. After observing t_i , each agent i sends a *message* $t'_i \in T$ to the principal. A message profile is a vector $t' = (t'_1, \dots, t'_n) \in T^n$.

Private communication. Private communication involves a *recommendation* from the principal to each agent, followed by the agent’s *report* to the mechanism. “Private” means that

recommendations are sent individually and remain off the record, so they are not subject to any constraints. Formally, after the formal rule $v^* \in M \subseteq \mathcal{C}$ is realized and a message profile $t' \in T^n$ is sent by the agents, the principal may send to each agent $i \in N$ a recommendation $\tau_i^{v^*}(t') \in T$. Since recommendations are private, agent i observes only $\tau_i^{v^*}(t')$.

After receiving a recommendation from the principal, each agent i sends a *report* $\tau'_i \in T$ to the mechanism. Finally, the outcome $v^*(\tau')$ is implemented, where a vector $\tau' = (\tau'_1, \dots, \tau'_n)$ represents a report profile.

Recommendation function. We introduce an auxiliary instrument to construct a mechanism in our model. For each formal rule v , we call a map $\rho^v: \mathcal{O} \rightarrow \mathcal{T}$ an (*outcome-based*) *recommendation function* for v if $v(\rho^v(o)) = o$ for each $o \in v(\mathcal{T})$. Thus, $\rho^v(o)$ specifies a report profile that induces outcome o under the formal rule v .⁴

Constraints on formal rules. As discussed above, formal rules are subject to certain constraints. We now outline the constraints considered in this paper.

Definition 1 (Constraints). The following properties are defined as constraints:

Symmetry: Rule $v \in \mathcal{R}$ is *symmetric* if for each permutation $\pi: N \rightarrow N$ and each $(t_1, \dots, t_n) \in \mathcal{T}$, $v(t_1, \dots, t_n) = (o_1, \dots, o_n)$ implies $v(t_{\pi(1)}, \dots, t_{\pi(n)}) = (o_{\pi(1)}, \dots, o_{\pi(n)})$.

Strict monotonicity: Suppose T is linearly ordered by $>$ and $\mathcal{O}$ is partially ordered by $>$.

Rule $v \in \mathcal{R}$ is *strictly monotone* if for each $t_i, t'_i \in T$ and $t_{-i} \in T^{n-1}$ with $t_i > t'_i$, $v(t_i, t_{-i}) > v(t'_i, t_{-i})$.

Continuity: Suppose (T, d_T) and $(\mathcal{O}, d_{\mathcal{O}})$ are metric spaces. Rule $v \in \mathcal{R}$ is *continuous* if it is continuous (with the product topology on T^n).

We will refer to symmetry as *fairness constraints*, as it restricts the formal rule across permutations of agents. By contrast, we will refer to strict monotonicity and continuity as *predictability constraints*, since they discipline the mapping from reports to outcomes.

⁴The outcome-based recommendation functions introduced above are auxiliary objects used for construction. In the definition of implementability presented in Section 5, we work directly with recommendation policies that map a message to recommended reports.

The timing of decisions The principal uses private communication and a *formal rule* $\nu^* \in \mathcal{C}$ to implement a target rule μ . In detail, the game proceeds in four stages:

Stage 1. The principal chooses a *menu* $M \subseteq \mathcal{C}$ and draws $\nu^* \in M$ according to a countably additive probability measure P on M . The drawn ν^* is hidden from the agents.

Stage 2. Each agent i observes type $t_i \in T$ and sends a *message* $t'_i \in T$ to the principal.

Stage 3. Observing ν^* and t' , the principal privately sends a *recommendation* $\tau_i^{\nu^*}(t') \in T$ to each agent i .

Stage 4. Each agent i sends a *report* $\tau'_i \in T$ to the institution,⁵ and the outcome $\nu^*(\tau')$ is implemented.

We refer to the tuple $(M, P, \{(\tau_i^\nu)_{i \in N}\}_{\nu \in M})$ as a *mechanism*, where M is a set of formal rules, P is a distribution over M , and for each $\nu \in M$, $(\tau_i^\nu)_{i \in N}$ is a profile of functions $\tau_i^\nu: T^n \rightarrow T$ that maps each report profile to a recommended report profile.

4. Mechanism

To construct a mechanism $(M, P, \{(\tau_i^\nu)_{i \in N}\}_{\nu \in M})$, we introduce an auxiliary *recommendation device* $(M, P, \{\rho^\nu\}_{\nu \in M})$, where for each $\nu \in M$, $\rho^\nu: \mathcal{O} \rightarrow \mathcal{T}$ is an outcome-based recommendation function. Given any mapping $\kappa: \mathcal{T} \rightarrow \mathcal{O}$, we construct a recommendation policy $\tau^\nu: \mathcal{T} \rightarrow \mathcal{T}$ by $\tau^\nu := \rho^\nu \circ \kappa$, and let τ_i^ν denote its i -th component, where $\kappa = \mu$ in Section 5. The auxiliary recommendation device $(M, P, \{\rho^\nu\}_{\nu \in M})$ then induces a mechanism $(M, P, \{(\tau_i^\nu)_{i \in N}\}_{\nu \in M})$ in the sense of Section 3. We say that $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfies *formal symmetry* if each $\nu \in M$ is symmetric.

$(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfies the *almost-sure obedience* property if, conditional on the agent's received recommendation, i.e., given that $\tau_i = \rho_i^\nu(o)$, for each alternative report $\tau'_i \neq \tau_i$, the probability that τ'_i yields a strictly better outcome is zero. The formal definition is as follows.

Definition 2 (Almost-sure obedience). $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfies the *almost-sure obedience* property if for each agent $i \in N$, $t \in \mathcal{T}$, $o \in \mathcal{O}$, each report profile $(\tau_i, \tau_{-i}) \in \mathcal{T} = T^n$, and each

⁵The institution has no strategic role; it merely applies the formal rule ν^* to the submitted reports.

alternative report $\tau'_i \in T$ with $\tau'_i \neq \tau_i$, if

$$\Pr_{\nu \sim P}[\tau_i = \rho_i^\nu(o)] > 0,$$

we have

$$\Pr_{\nu \sim P}[u_i(t, \nu(\tau'_i, \tau_{-i})) > u_i(t, \nu(\tau_i, \tau_{-i})) \mid \tau_i = \rho_i^\nu(o)] = 0.$$

We show that there exists $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfying the almost-sure obedience property and formal symmetry.⁶

Lemma 1. *There exists a recommendation device $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfying almost-sure obedience, formal symmetry, and $\nu(\mathcal{T}) \supseteq \mathcal{O}^*$ for each $\nu \in M$.*

Let us explain in detail how to construct a mechanism by specifying a family of admissible formal rules and associated recommendation functions. Fix a symmetric rule $\nu \in \mathcal{C}$. For a target outcome $o \in \mathcal{O}^*$, let $\tau^* \in \mathcal{T}$ be such that $\nu(\tau^*) = o$. Since T is a standard Borel space with $|T| = |\mathbb{R}| = |\mathbb{R}^2|$, there exist Borel bijections $g: (0, 1) \rightarrow T$ and $h: T \rightarrow (0, 1)^2$. The bijection $h: T \rightarrow (0, 1)^2$ encodes each individual report $\tau_i \in T$ as a pair (s_i, w_i) where $s_i \in (0, 1)$ identifies which report is being used, and $w_i \in (0, 1)$ serves as an auxiliary label that will be used for verification. Let $\varepsilon = (\varepsilon_s)_{s \in (0, 1)}$, where for each $s \in (0, 1)$, an auxiliary random variable ε_s is independently drawn according to the uniform distribution on $(0, 1)$. These auxiliary random variables provide a continuum of independent labels, indexed by the identifying component s .

Given a realization of ε , define a new formal rule ν^ε as follows. Consider each report profile τ , and write $h(\tau_i) = (s_i, w_i)$ for each agent i . The formal rule ν^ε checks whether the verification condition, $w_i = \varepsilon_{s_i}$, for all i , is satisfied. If this condition holds for all agents, the rule implements $\nu(g(s_1), \dots, g(s_n))$, that is, it applies the original rule ν to the reports identified by the components $(s_1, \dots, s_n)$. If the condition fails for exactly one agent i , the rule implements the individual worst outcome $\underline{o}^i$; if it fails for two or more agents, the rule implements the symmetric fallback outcome o^* , as specified in Assumption 1.

⁶All proofs of the lemma and propositions are relegated to the Appendix.

For the target outcome o , the principal recommends to agent i the report $\rho_i^{v^e}(o) = h^{-1}(s_i^*, \varepsilon_{s_i^*})$, where $h(\tau_i^*) = (s_i^*, w_i^*)$. That is, the recommendation instructs agent i to use the report whose identifying component coincides with that of the recommended report τ_i^* and whose verification label matches the corresponding auxiliary random variable.

Because the auxiliary random variables $(\varepsilon_s)_{s \in (0,1)}$ are generated independently of agents' identities, the construction treats all agents symmetrically, and hence each v^e is a symmetric formal rule. Moreover, since the verification labels are drawn from a continuum, any report that differs from the recommended one satisfies the verification condition with probability zero. As a result, no agent can improve upon the recommended behavior by deviating.

5. Implementation

5.1. Implementation with fairness constraints

Building on $(M, P, \{\rho^v\}_{v \in M})$ found in lemma 1, we now present our implementation results. First, by using $(M, P, \{\rho^v\}_{v \in M})$ and a given rule μ , we can construct a recommendation policy $\tau^v = \rho^v \circ \mu$, which induces a mechanism $(M, P, \{\tau^v\}_{v \in M})$. We consider whether μ is implementable by a mechanism $(M, P, \{\rho^v \circ \mu\}_{v \in M})$. To state the conditions for implementation, we first recall standard notions of incentive compatibility.

Definition 3. A rule μ is *dominant-strategy incentive compatible (DSIC)* if, for each $i \in N$, $t_{-i} \in T^{n-1}$, and $t_i, t'_i \in T$,

$$u_i((t_i, t_{-i}), \mu(t_i, t_{-i})) \geq u_i((t_i, t_{-i}), \mu(t'_i, t_{-i})),$$

and strict inequality holds for at least one $t'_i \in T$.

We say that a rule $\mu \in \mathcal{R}$ is *implementable* via $(M, P, \{\rho^v\}_{v \in M})$ ⁷ under sequential dominant-strategy if sending true type to the principal and reporting the recommended type to the institution are optimal for the agents in the dominant-strategy sense. More precisely:

⁷Standard implementation theory discusses whether a rule is implemented by a *mechanism*. In this study, since the mechanism is naturally induced by a recommendation device $(M, P, \{\rho^v\}_{v \in M})$, we state implementability in terms of the recommendation device rather than the mechanism itself.

Definition 4. A rule $\mu \in \mathcal{R}$ is *implementable* via $(M, P, \{\rho^v\}_{v \in M})$ under sequential dominant-strategy if the following conditions hold:

(a) For each $v \in M$, $i \in N$, $t_{-i} \in T^{n-1}$, and $t_i, t'_i \in T$,

$$u_i(t, v \circ \rho^v \circ \mu(t)) \geq u_i(t, v \circ \rho^v \circ \mu(t'_i, t_{-i})).$$

(b) For each $i \in N$, $t \in \mathcal{T}$, $\tau'_i \in T$, and $\tau_{-i} \in T^{n-1}$,

$$\mathbb{E}_{v \sim P}[u_i(t, v(\rho_i^v \circ \mu(t), \tau_{-i})) \mid \tau_i = \rho_i^v \circ \mu(t)] \geq \mathbb{E}_{v \sim P}[u_i(t, v(\tau'_i, \tau_{-i})) \mid \tau_i = \rho_i^v \circ \mu(t)].$$

Moreover, strict inequality holds for some t'_i in (a) and for some τ'_i in (b).

In our notion of implementability, agents are required to play a (weakly) dominant strategy in the communication stage, taking truthful reporting as given. We adopt the definition of dominance in [Azrieli and Jain \(2018\)](#), but we impose it only on the communication stage. By contrast, [Azrieli and Jain \(2018\)](#) impose dominance and symmetry on the *entire* mechanism, thereby restricting implementation to symmetric rules.⁸

With these notions in hand, we state a sufficient condition for implementation when constraints are imposed only on the auditable formal rule. The next result follows immediately from the almost-sure obedience property (Definition 2).

Proposition 1. *Suppose that μ is DSIC and $\mu(\mathcal{T}) \subseteq \mathcal{O}^*$. Then, μ is implementable under sequential dominant-strategy via $(M, P, \{\rho^v\}_{v \in M})$ that satisfies formal symmetry.*

Proposition 1 shows that a fairness constraint such as symmetry does not bite: imposing symmetry on the formal rule alone does not restrict implementability once private recommendations are allowed. Proposition 1 is an immediate consequence of our sufficient condition. Since μ is DSIC and takes values in $\mathcal{O}^*$, Lemma 1 constructs $(M, P, \{\rho^v\}_{v \in M})$ that satisfies the almost-sure obedience property. Applying the almost-sure obedience property and DSIC of μ

⁸As noted in section 2, [Chen and Knyazev \(2023\)](#) show that under a weaker notion of dominance (M-DSE), some asymmetric rules can be implemented by symmetric mechanisms.

yields sequential dominant-strategy implementability, and $(M, P, \{\rho^v\}_{v \in M})$ satisfies formal symmetry because each $v \in M$ is symmetric.⁹

5.2. Implementation with predictability constraints

This subsection studies predictability constraints, focusing on strict monotonicity and continuity. Unlike fairness constraints, these predictability constraints are incompatible with the almost-sure obedience property and thus retain bite for implementability even when private communication is allowed.

The intuition is straightforward. If rules are required to be strictly monotone, then once agent i knows that the recommended report τ_i induces outcome $o \in \mathcal{O}^*$, any higher report $\tau'_i > \tau_i$ induces a different outcome $o' > o$ with probability one; if the agent prefers o' to o , the almost-sure obedience property fails.

Regarding continuity, if, for some $v^* \in M$, the deviation (τ'_i, τ_{-i}) yields an outcome that is different from $v^*(\tau)$ and is not equal to any worst outcome, then by continuity, v^* also maps a neighborhood of (τ'_i, τ_{-i}) to non-worst outcomes. Hence, under any probability measure over M , the conditional probability of a strict improvement after deviating is positive, contradicting the almost-sure obedience property. The results are summarized as follows:

Proposition 2. *Consider an arbitrary recommendation device $(M, P, \{\rho^v\}_{v \in M})$.*

- (1) *If each $v \in M$ is monotone, $(M, P, \{\rho^v\}_{v \in M})$ fails to satisfy almost-sure obedience for some utility profile.*
- (2) *Suppose T is separable and each $v \in M$ is continuous. Then, $(M, P, \{\rho^v\}_{v \in M})$ fails to satisfy almost-sure obedience for some continuous utility profile.*

Proposition 2 formalizes that predictability constraints retain real bite: requiring the formal rules to be strictly monotone or continuous is incompatible with almost-sure obedience and therefore cannot, in general, be rendered ineffective by private recommendations. This stands in sharp contrast to Proposition 1, where imposing symmetry only on the formal rule no longer

⁹When the incentive-compatibility property of μ is weakened to be Bayesian incentive compatible, the corresponding conclusion of Proposition 1 also holds: μ is implementable under a perfect Bayesian Nash equilibrium.

restricts implementability once private communication is allowed. In sum, fairness constraints restrict how the rule treats agents across permutations, whereas predictability constraints discipline how outcomes must respond to changes in reports; it is this disciplining feature that prevents the “punishment” structures needed for almost-sure obedience.

6. Extensions and discussion

This section provides extensions and further discussion. We consider approximate versions of the obedience requirement and address additional modeling issues, including continuity, countable type spaces, limited commitment, and coalitional deviations.

Approximate obedience. We relax the almost-sure obedience property by allowing a small probability that some deviation from the recommended report is profitable.

Definition 5. Fix $\eta > 0$. $(M, P, \{\rho^v\}_{v \in M})$ satisfies the η -obedience property if, conditional on receiving the recommended message, the probability that some unrecommended report yields a *strictly* better outcome (relative to Definition 2) is strictly less than η .

Even under the η -obedience property alone, some rules may fail to be implementable.

Example 2. Let $n = 2$ and $\mathcal{O} = \{(1, 0), (1/2, 1/2), (0, 1)\}$. Agent 2 prefers o to o' iff $o_2 > o'_2$, while agent 1 is indifferent across all outcomes. Take worst outcomes $\underline{o}^1 = (0, 1)$, $\underline{o}^2 = (1, 0)$ and symmetric fallback $o^* = (1/2, 1/2)$. Consider the (trivially DSIC) rule $\mu(t) \equiv (1, 0)$ for all $t \in \mathcal{T}$. Suppose a symmetric formal rule ν^* implements μ . Then there is $\tau \in \mathcal{T}$ with $\nu^*(\tau) = (1, 0)$, and by symmetry $\nu^*(\tau_1, \tau_1) = (1/2, 1/2)$. Since agent 2 never prefers $(1, 0)$ to $(1/2, 1/2)$, any deviation with $\tau'_2 = \tau_1$ delivers $(1/2, 1/2)$ with positive probability. Even when this probability is sufficiently small, μ is not implementable, as the deviation yields a strict improvement in expectation.

A sufficient way to preclude this problem is to ensure a strict gap between on-path outcomes and each agent’s worst outcome. Suppose that there exists $\alpha > 0$ such that, for each $i \in N$, each $t \in \mathcal{T}$, and each $o \in \mu(\mathcal{T})$,

$$u_i(t, o) \geq u_i(t, \underline{o}^i) + \alpha.$$

If μ satisfies this property, then μ remains implementable even when $(M, P, \{\rho^v\}_{v \in M})$ satisfies only the approximate obedience property (for sufficiently small η): the potential gain from any deviation that occasionally hits a non-worst outcome is dominated by the on-path gap α .

This concern is especially salient in assignment and matching problems. With indivisible objects and no transfers, “receiving nothing” is both an agent’s worst outcome and an outcome that may arise under *any* rule when supply is scarce; similarly, in matching, “no match” is individually rational and may occur on a path whenever market sides are unbalanced. In such environments, the strict gap condition typically fails, and the approximate obedience property alone will not achieve implementability.

Continuity. As shown in Proposition 2, the *exact* obedience property is incompatible with continuity. Nevertheless, continuous rules can satisfy the *approximate* obedience property.

Proposition 3. *Fix $\eta > 0$. Suppose that $\mathcal{O}$ is convex, $T = \mathbb{R}$, and the symmetric worst outcomes are common (i.e., there exists $\underline{o} \in \mathcal{O}$ such that $\underline{o}^i = \underline{o}$ for each $i \in N$). Let $\mathcal{O}^* \subseteq \mathcal{O}$ be a countable set of outcomes. Then there exists a symmetric menu M^* of continuous rules that are surjective onto $\mathcal{O}^*$ and a collection $\{\rho^v\}_{v \in M^*}$ such that $(M^*, P, \{\rho^v\}_{v \in M^*})$ satisfies the η -obedience property.*

A separate question concerns exact obedience. Our impossibility result (Proposition 2(2)) relies on constructing a utility profile in which the target outcome is a local common worst outcome. When outcomes are required to be Pareto efficient, such a construction is no longer available, and the question of whether exact obedience is compatible with continuity remains open. In particular, it may be possible to design continuous rules that satisfy exact almost-sure obedience by exploiting the conflicting preferences inherent in Pareto efficient outcomes—penalizing each agent’s deviation along a direction unfavorable to that agent. We leave this as an open question.

Countable types. When the type space is countable, exact almost-sure obedience cannot be achieved. Indeed, suppose that for some profile $\tau \in \mathcal{T}$ and deviation $\tau'_i \in T$ we have, for each $v \in \mathcal{C}$, $v(\tau) \neq v(\tau'_i, \tau_{-i}) \neq \underline{o}^i$. By countability of T and countable additivity of P over M , there

exists τ'_i such that

$$\Pr_{v \sim P} \left(u_i(t, v(\tau'_i, \tau_{-i})) > u_i(t, v(\tau_i, \tau_{-i})) \mid \tau_i = \rho_i^v(o) \right) > 0.$$

In words, an agent can “guess” a password with strictly positive probability.

By contrast, for each $\eta > 0$, one can construct a mechanism that is η -obedient. The construction parallels Lemma 1 but draws passwords from a finite grid: instead of sampling from $(0, 1)$, pick passwords from $(0, 1) \cap \{1/m, 2/m, \dots, m/m\}$ for some large $m \in \mathbb{N}$. Since $\mathcal{T}$ is countable, there is a bijection $\mathcal{T} \rightarrow ((0, 1) \cap \mathbb{Q}) \times \{1/m, 2/m, \dots, m/m\}$, which allows us to encode each type together with a password. Taking m sufficiently large makes the probability of a successful, unrecommended deviation arbitrarily small, yielding an η -obedient analogue of Lemma 1.

Incentive of the principal. Our analysis assumes that the principal can commit to the recommendation policy. This assumption highlights what can and cannot be achieved when regulation focuses solely on the externally auditable formal rule, while off-the-record recommendations remain beyond the auditor’s reach. A natural direction for future work is to explore environments with limited commitment or explicit restrictions on off-the-record instruments.¹⁰

As a simple benchmark for the other extreme, consider the case in which the principal cannot commit to the recommendation policy $\tau^v: \mathcal{T} \rightarrow \mathcal{T}$ at all.¹¹ Then, the rule μ should satisfy ex post optimality to ensure truthful recommendations by the principal; that is, $u_0(t, \mu(t)) \geq u_0(t, o)$ for each $o \in \mathcal{O}^*$ and each $t \in \mathcal{T}$. Conversely, if μ is ex post optimal and incentive compatible, then some $v \in \mathcal{C}$ implements μ in a perfect Bayesian (Nash) equilibrium of the full game.

Coalitional deviations. Beyond individual deviations, coalitions can in principle manipulate the formal rule. Because the password test in our mechanism is applied *coordinate-wise* and the same password map $\varepsilon: (0, 1) \rightarrow (0, 1)$ is used for every coordinate, a coalition can

¹⁰We thank an anonymous referee for suggesting this direction. In a related vein, [Grigoryan and Möller \(2024\)](#) study when a group of agents can audit the mechanism and detect the principal’s deviation, thereby disciplining the principal’s behavior.

¹¹See, for example, [Baliga, Corchon, and Sjöström \(1997\)](#) and [Bester and Strausz \(2000\)](#) for mechanism design without commitment.

permute the recommended pairs and mimic the recommendations for others without triggering any punishment. Formally, fix $v^{\varepsilon} \in M$, let $o \in \mathcal{O}$ be the target outcome, and let $\tau = \rho^{v^{\varepsilon}}(o) \in \mathcal{T}$ be the recommended report profile.

Consider $\tau' \in \{\tau_1, \dots, \tau_n\}^n$. Since each coordinate (s_i, w_i) of $h(\tau')$ still satisfies $w_i = \varepsilon_{s_i}$, the password test passes at every coordinate and no punishments are triggered; the set of achievable outcomes is extended from $v^{\varepsilon}(\tau)$ to $\{v^{\varepsilon}(\tau') \mid \tau' \in \{\tau_1, \dots, \tau_n\}^n\}$. By symmetry, this set includes all permuted outcomes of o : $(o_{\pi(1)}, \dots, o_{\pi(n)})$ for each permutation $\pi: N \rightarrow N$. Blocking such coalitional deviations would require additional structure beyond constraints on the auditable formal rule.

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A. Proofs

Proof of Lemma 1. Step 1: Construction of a subfamily of symmetric formal rules.

Fix a symmetric rule $v \in \mathcal{R}$ such that $v(\mathcal{T}) \supseteq \mathcal{O}^*$.¹² Since T has the same cardinality as $\mathbb{R}$, we have $|T| = |(0, 1)^2|$. Hence, there exist Borel bijections $g: (0, 1) \rightarrow T$ and $h: T \rightarrow (0, 1)^2$. Let $\Omega := (0, 1)^{(0, 1)}$ and take an arbitrary $\varepsilon \in \Omega$.

We define v^ε as follows: For each $\tau_i \in T$, let $(\tau_i^*, w_i) = h(\tau_i)$.

1. If $w_i = \varepsilon_{\tau_i^*}$ for each $i \in N$, then $v^\varepsilon(\tau) = v(g(\tau_1^*), \dots, g(\tau_n^*))$.
2. If there is i such that $w_i \neq \varepsilon_{\tau_i^*}$ and $w_j = \varepsilon_{\tau_j^*}$ for each $j \in N \setminus \{i\}$, then $v^\varepsilon(\tau) = \underline{o}^i$.
3. Otherwise, $v^\varepsilon(\tau) = o^*$.

¹²Since T has the same cardinality as $\mathbb{R}$, we can construct a symmetric v . Let $f(s_1, \dots, s_n) = s_1 + \dots + s_n$ for each $s_i \in \mathbb{R}$. This is a symmetric and surjective function on $\mathbb{R}$. Let $\psi: T \rightarrow \mathbb{R}$ be a Borel bijection. Then, as there is a surjection $\varphi: \mathbb{R} \rightarrow \mathcal{O}$, let $v(t) = \varphi(f(\psi(t_1), \dots, \psi(t_n)))$, which is a symmetric function with $v(\mathcal{T}) \supseteq \mathcal{O} \supseteq \mathcal{O}^*$.

We now show that each ν^ε is symmetric. Let π be any permutation of N and fix $\tau \in T^n$. Write $h(\tau_i) = (\tau_i^*, w_i)$ for each $i \in N$.

If $w_i = \varepsilon_{\tau_i^*}$ for every i , then

$$\nu^\varepsilon(\tau) = \nu(g(\tau_1^*), \dots, g(\tau_n^*)) \quad \text{and} \quad \nu^\varepsilon(\tau_\pi) = \nu(g(\tau_{\pi(1)}^*), \dots, g(\tau_{\pi(n)}^*)),$$

which are equal by the symmetry of ν .

If there exists a unique $i \in N$ such that $w_i \neq \varepsilon_{\tau_i^*}$ and $w_j = \varepsilon_{\tau_j^*}$ for all $j \in N \setminus \{i\}$, then by construction $\nu^\varepsilon(\tau) = \underline{o}^i$. Under the permutation π , the unique violator becomes $\pi(i)$, and thus $\nu^\varepsilon(\tau_\pi) = \underline{o}^{\pi(i)}$, which is consistent with symmetry.

Otherwise, at least two agents violate $w_j = \varepsilon_{\tau_j^*}$ and hence $\nu^\varepsilon(\tau) = o^* = \nu^\varepsilon(\tau_\pi)$. Since o^* is symmetric, ν^ε is symmetric.

Step 2: Construction of the probability space.

Let $((0, 1), \mathcal{B}, \lambda)$ denote the unit interval with its Borel σ -algebra and Lebesgue measure. Define

$$\mathcal{F} := \sigma(\{\varepsilon \in \Omega \mid \varepsilon_{x_1} \in (0, 1), \dots, \varepsilon_{x_\ell} \in (0, 1), \ell \in \mathbb{N}\}),$$

where $\sigma(X)$ denotes the σ -algebra generated by a set X . Let P be the (product) probability measure on $(\Omega, \mathcal{F})$ such that for every finite set $\{x_1, \dots, x_\ell\} \subseteq (0, 1)$, the random vector $(\varepsilon_{x_1}, \dots, \varepsilon_{x_\ell})$ is i.i.d. with common law λ (uniform distribution on $(0, 1)$).¹³ We write $\varepsilon_x(\omega)$ for the coordinate map at index $x \in (0, 1)$.

Let $f: \Omega \rightarrow M_\nu$ be the bijection $\varepsilon \mapsto \nu^\varepsilon$ and endow M_ν with the transported σ -algebra $\mathcal{M} := \{A \subseteq M_\nu \mid f^{-1}(A) \in \mathcal{F}\}$. Define $P_{M_\nu}(A) := P(f^{-1}(A))$ for each $A \in \mathcal{M}$.

Step 3: Construction of the recommendation functions.

For each $o \in \mathcal{O}$ such that $o = \nu^\varepsilon(t) = \nu(g(t_1^*), \dots, g(t_n^*))$, let $\rho_i^{\nu^\varepsilon}(o) = h^{-1}(t_i^*, \varepsilon_{t_i^*})$ for each $i \in N$. As ν is surjective, this is well-defined.

Step 4: Proof of the almost-sure obedience property.

¹³By the Kolmogorov (Daniell–Kolmogorov) extension theorem, the consistent family of finite-dimensional distributions $\{\lambda^{\otimes \ell}\}_{\ell \in \mathbb{N}}$ determines a *unique* probability measure P on the product measurable space $(\Omega, \mathcal{F})$.

Now we show that $(P, M_\nu, \{\rho^{\nu'}\}_{\nu' \in M_\nu})$ satisfies the almost-sure obedience property.

For each $o \in \mathcal{O}$, by $\tau_i = \rho_i^{\nu^e}(o)$, $h(\tau_i) = (x, \varepsilon_x)$ for some $x \in (0, 1)$. For each deviation $\tau'_i \neq \tau_i$, let $(x', w') = h(\tau'_i)$. Then, the probability that $w' = \varepsilon_{x'}$ is 0. We have three cases with respect to the others' reporting.

Case 1. Suppose that $\tau_j = \rho_j^{\nu^e}(o)$ for each $j \in N \setminus \{i\}$. Then, $\nu^e(\tau'_i, \tau_{-i}) = \underline{o}^i$ with probability 1. Thus, no deviation is profitable.

Case 2. Suppose that $\left| \left\{ j \in N \setminus \{i\} \mid \tau_j \neq \rho_j^{\nu^e}(o) \right\} \right| = 1$. Note that if $\tau_j \neq \rho_j^{\nu^e}(o)$, with probability 1, $w_j \neq \varepsilon_{\tau_j^*}$, where $(\tau_j^*, w_j) = h(\tau_j)$. Then, $\nu^e(\tau'_i, \tau_{-i}) = o^*$ with probability 1, while $\nu^e(\tau_i, \tau_{-i}) = \underline{o}^j$. Thus, no deviation is profitable.

Case 3. Suppose that $\left| \left\{ j \in N \setminus \{i\} \mid \tau_j \neq \rho_j^{\nu^e}(o) \right\} \right| \geq 2$. Then, no deviation changes the result. $\square$

Proof of Proposition 1. Consider $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfying the almost-sure obedience property constructed in Lemma 1.

For each $\nu \in M$, let $\tau^\nu: \mathcal{T} \rightarrow \mathcal{T}$ satisfy $\tau^\nu(t) = \rho^\nu(\mu(t))$, and let $\tau_i = \tau_i^\nu(t)$.

Consider stage 4. Suppose the message sent by the principal to agent i is τ_i^* . Then, agent i believes that sending τ_i yields $\nu(\tau_i, \tau_{-i})$ when the others' report is τ_{-i} . By the almost-sure obedience property, agents have no incentive to misreport $\tau_i \neq \tau_i^*$. Furthermore, if $\tau_{-i} = \tau_{-i}^*$, the outcome is $\mu(t)$ when $\tau_i = \tau_i^*$.

Consider stage 2. Conditional on the equilibrium strategy of the agents in stage 4, reporting t_i to the principal yields outcome $\mu(t_i, t_{-i})$. Then, by the incentive compatibility of μ , truth-telling is the dominant strategy.

Furthermore, by the dominant-strategy incentive compatibility of μ , the optimality is strict for at least one t'_i : $\mu(t) > \mu(t'_i, t_{-i})$. $\square$

Proof of Proposition 2. Proof of (1): Consider a $(M, P, \{\rho^\nu\}_{\nu \in M})$ for strictly monotonic formal rules.

Consider a utility function for agent $i \in N$ such that $u_i(t, o) > u_i(t, o')$ whenever $o > o'$.

Let $\nu \in M$ be a strictly monotone rule. Suppose that $\nu(\tau) = o$ for some $\tau \in \mathcal{T}$, and consider a deviation $\tau'_i > \tau_i$. Then, the deviation $\tau'_i > \tau_i$ improves the utility with probability 1, and therefore the almost-sure obedience property fails.

Proof of (2): Consider a $(M, P, \{\rho^\nu\}_{\nu \in M})$ for continuous formal rules.

Fix $o' \in \mathcal{O}^*$. Consider a continuous utility function such that $u_i(t, \hat{o}) > u_i(t, o')$ for each $i \in N$ and each $\hat{o} \in B_\eta(o') \setminus \{o'\}$ for some $\eta > 0$. In other words, o' is a local common worst outcome.

Suppose that each $v \in M$ is continuous and $v(\mathcal{T}) \supseteq \mathcal{O}^*$. For each $v \in M$, since $v(\mathcal{T}) \supseteq \mathcal{O}^*$, there exist $\tau, \tau' \in \mathcal{T}$ such that $v(\tau) \neq v(\tau') = o'$. By changing coordinates one at a time, there exists $i(v) \in N$ such that $v(a^v, b^v) \neq o' = v(c^v, b^v)$, where a^v and c^v denote the $i(v)$ -th coordinates of the profiles that differ, and $b^v \in T^{n-1}$ denotes the remaining coordinates (constructed by setting $\tau_j'' = \tau_j$ for $j < i(v)$ and $\tau_j'' = \tau_j'$ for $j > i(v)$). We may choose $c^v = \rho_{i(v)}^v(o')$ by setting $\tau' = \rho^v(o')$.

By taking a^v sufficiently close to c^v (using the freedom in the choice of τ), continuity of v ensures $v(a^v, b^v) \in B_\eta(o') \setminus \{o'\}$, so that

$$u_i(t, v(a^v, b^v)) > u_i(t, o') = u_i(t, v(c^v, b^v)).$$

Since v and u_i are continuous, there exist open sets $U_1(v) \ni a^v$ in T , $U_2(v) \ni b^v$ in T^{n-1} , and $U_3(v) \ni c^v$ in T such that

$$u_i(t, v(\hat{a}, \hat{b})) > u_i(t, v(\hat{c}, \hat{b})) \quad \text{for all } (\hat{a}, \hat{b}, \hat{c}) \in U_1(v) \times U_2(v) \times U_3(v).$$

As T is separable, so is T^{n-1} . Let $D = \{d_1, d_2, \dots\}$ be a countable dense subset of T and $D' = \{d'_1, d'_2, \dots\}$ a countable dense subset of T^{n-1} . Define

$$M_{k,l,m}^i = \{v \in M \mid i(v) = i, d_k \in U_1(v), d'_l \in U_2(v), d_m \in U_3(v)\}.$$

Then $\bigcup_{i \in N} \bigcup_{k,l,m \in \mathbb{N}} M_{k,l,m}^i = M$. By countable additivity of P , $P(M_{k,l,m}^i) > 0$ for some (i, k, l, m) .

For every $v \in M_{k,l,m}^i$, since $(d_k, d'_l, d_m) \in U_1(v) \times U_2(v) \times U_3(v)$,

$$u_i(t, v(d_k, d'_l)) > u_i(t, v(d_m, d'_l)).$$

Moreover, since $\rho_i^v(o') \in U_3(v)$, we also have

$$u_i(t, v(d_k, d'_l)) > u_i(t, v(\rho_i^v(o'), d'_l)).$$

Thus, for every $\nu \in M_{k,l,m}^i$, the deviation from the recommendation $\rho_i^\nu(o')$ to d_k with $\tau_{-i} = d'_l$ yields a strict improvement. Since $P(M_{k,l,m}^i) > 0$, this contradicts the almost-sure obedience property. $\square$

Proof of Proposition 3. Let $T^* = \mathbb{N}$, which is a countable subset of T . Then, there is a surjection $\nu^*: (T^*)^n \rightarrow \mathcal{O}^*$. Let $\varepsilon \in \Omega := (\frac{1}{3}, \frac{2}{3})^{\mathbb{N}}$. For each $\tau_i \in \mathbb{R}$, let $k_i = \lfloor \tau_i \rfloor$ and $w_i = \tau_i - \lfloor \tau_i \rfloor$ for each $i \in N$.

Take real numbers $\beta = (\beta_1, \beta_0)$ with $1 > \beta_1 > \beta_0 > 0$, and define $q: (0, 1) \times \mathbb{N} \rightarrow (0, 1)$ as

$$q(w, k) = \begin{cases} 0 & \text{if } |w - \varepsilon_k| \leq \beta_0 \\ \frac{|w - \varepsilon_k| - \beta_0}{\beta_1 - \beta_0} & \text{if } \beta_0 < |w - \varepsilon_k| < \beta_1 \\ 1 & \text{if } |w - \varepsilon_k| \geq \beta_1, \end{cases}$$

and

$$Q^{\varepsilon, \beta}(t) = 1 - \prod_{i=1}^n (1 - q(w_i, k_i)).$$

Then, we define $\nu^{\varepsilon, \beta}$ as follows:

$$\nu^{\varepsilon, \beta}(\tau) = Q^{\varepsilon, \beta}(\tau) \cdot \underline{o} + [1 - Q^{\varepsilon, \beta}(\tau)] \cdot \nu^*(k_1, \dots, k_n).$$

This implies that if $|w_i - \varepsilon_{\tau_i}| > \beta_1$, then $\nu^{\varepsilon, \beta}(\tau) = \underline{o}$. Therefore, as $\beta_1 < 1/3$, $\nu^{\varepsilon, \beta}$ is continuous. Let $\rho^{\nu^{\varepsilon, \beta}}(o)$ be a recommendation function such that $\nu^*(k_1, \dots, k_n) = o$ and $\rho_i^{\nu^{\varepsilon, \beta}}(o) = k_i + \varepsilon_{k_i}$ for each $i \in N$.

We independently choose $\varepsilon_k \in (1/3, 2/3)$ from the uniform distribution on $(1/3, 2/3)$ for each $k \in \mathbb{N}$. Let $z \in (0, 1/3)$ and $\ell \in \mathbb{N}$. We also choose β_1 from $\{z/\ell, 2z/\ell, \dots, z\}$ with equal probability, and set $\beta_0 = \beta_1 - 1/\ell$. Let $M = \{\nu^{\varepsilon, \beta} \mid \varepsilon \in (1/3, 2/3)^{\mathbb{N}}, \beta_1 \in \{z/\ell, 2z/\ell, \dots, z\}\}$. Let (M, P) denote such a probability space.

Next, we check the deviation incentive. Let $\tau_i = \rho_i^{\nu^{\varepsilon, \beta}}(o)$, and $\tau'_i \neq \tau_i$. Let $k_i = \lfloor \tau_i \rfloor$, $w_i = \tau_i - k_i$, $k'_i = \lfloor \tau'_i \rfloor$, and $w'_i = \tau'_i - k'_i$. Consider $t'_i \in \mathbb{R}$ with $k'_i \neq k_i$. Then, the probability that $|w'_i - \varepsilon_{k'_i}| < \beta_1$ is at most $6z$.

Consider $t'_i \in \mathbb{R}$ with $k_i = k'_i$. Then, as the agent knows $\varepsilon_{k'_i}$, we consider the probability that $\beta_0 < |w'_i - \varepsilon_{k'_i}| < \beta_1$. As β_1 is randomly chosen, this probability is at most $\frac{1}{\ell}$.

By the same argument, the probability that the others' unrecommended reports change the result from $\{\nu^{\varepsilon, \beta}(\tau), \underline{o}\}$ is at most $\max\{6z, 1/\ell\}$. Therefore, the probability that $\nu^{\varepsilon, \beta}(\tau'_i, \tau_{-i}) \notin \{\underline{o}, \nu^{\varepsilon, \beta}(\tau)\}$ is at most $1 - [1 - \max\{6z, 1/\ell\}]^n$. In conclusion, by taking z sufficiently small and ℓ sufficiently large, the probability of strict improvement can be made arbitrarily small, and therefore $(M, P, \{\rho^\nu\}_{\nu \in M})$ satisfies the η -obedience property.

Furthermore, if ν^* is symmetric, then $\nu^{\varepsilon, \beta}$ is also symmetric, as q is independent of agents' identities. □